Music Minimal Skip Optimizer

STAT 4830 Slides Week 7

Problem Statement

Skipping Songs

Skip behavior is core engagement sign for music platforms.

Skips can indicate:

- Shorter listening sessions
- Lower satisfaction
- Reduced retention

Objective

Given a fixed set of candidate tracks, find the best permutation to minimize cumulative predicted skip probability

Week 6 focus: optimize sequence quality in later session positions using a trained skip model

Impact

Ordering is a real product decision (Daily Mix, Radio, Discover Weekly)

Better ordering improves listening flow without changing the underlying songs.

Constraints and Risks

Constraints (Week 6)

- Model depends on richer features; preprocessing is heavier
- Tracks mostly independently treated when scoring, so transition quality not fully optimized
- Search still greedy, so can't guarantee globally optimal sequence order
- Validation remains offline (session-based split), so no direct causal proof from user behavior.

Risks

- Sigmoid outputs may be overconfident/underconfident for optimization use
- Generalization risk: gains measured on mini-subset may not fully transfer to broader catalog or user segments
- Strong reliance on hist_user_behavior_reason_start may reduce robustness in less explicit sessions

Data Updates

Spotify Sequential Skip Prediction (WSDM Cup 2018)

Data Updates

- Merged Spotify session logs + track metadata/audio features
- 167,880 merged rows
- 8,000 train / 2,000 test sessions

Feature Engineering

- Where the user is in the session and time context (*session_progress*, day/time/weekend)
- How smooth the transition is (*delta_**, acoustic similarity)
- Cumulative user behavior so far (*cum_skips*, *skip_rate_so_far*, pauses, tempo deviation)

Why it's suitable

- Moved from mostly heuristic signals to a context-aware feature space

Model and Optimization Update

Skip Prediction Model

- PyTorch logistic-regression model using sigmoid function
- Trained with BCE loss and Adam on session-based train/test split
- Produces per-track skip probabilities used as optimization costs

Reordering Update

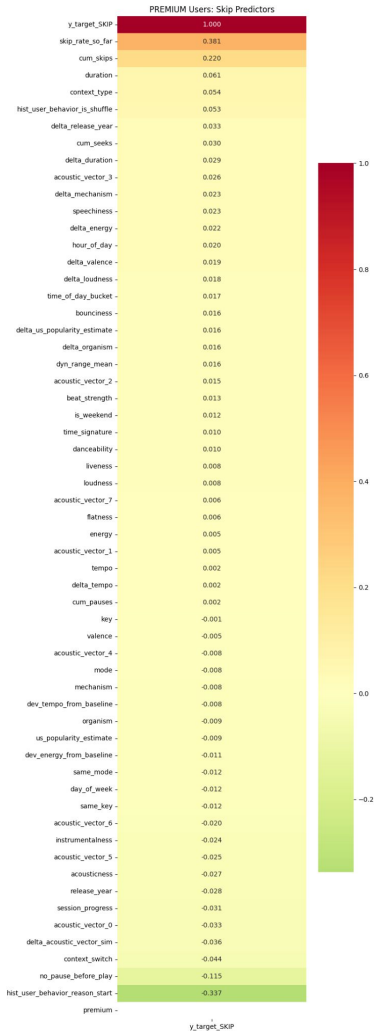
- Implemented greedy sequencing for playlist optimization
- At each step, select the remaining track with the lowest predicted skip probability
- Objective is to minimize total predicted skip probability over reordered sequence

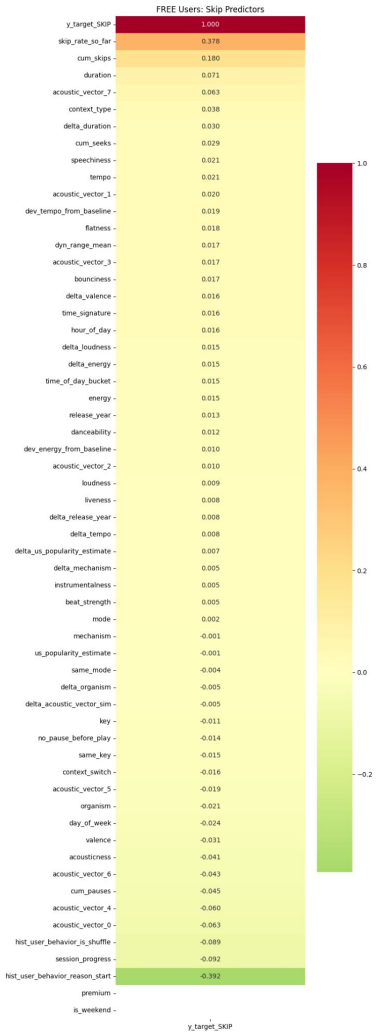
What changed from before

- Moved from heuristic frequency scores to a trainable model-driven optimization loop
- End-to-end pipeline now runs from feature engineering to sequence output

Week 6 Results

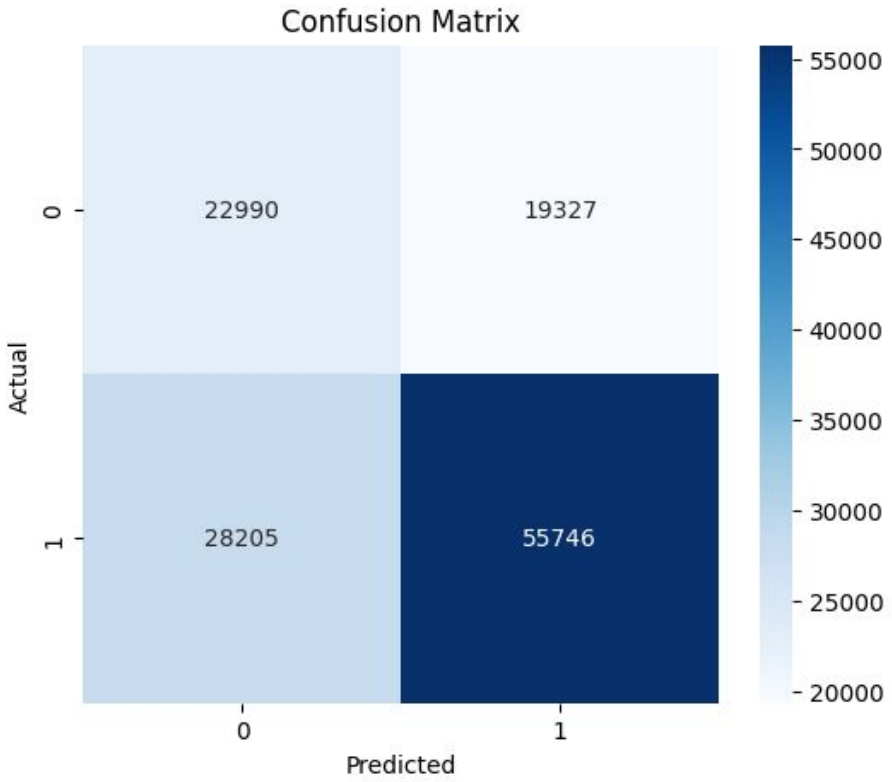
- Top skip drivers in both Premium and Free: *skip_rate_so_far* and *cum_skips*
    - *cum_skips* positive but *skip_rate_so_far* can weaken/flip due to session-position effects and skip constraints that free users have
- Session behavior/history matters more than individual audio features
- *hist_user_behavior_reason_start* is the strongest negative predictor
- Premium vs Free differences are negligible; overall pattern is consistent across groups





Week 6 Results

- The skip precision is 75%, and the recall is 66%.
- This indicates that the model is filtering slightly aggressively, with 19k false positives.
- The AUC-ROC is 0.64, suggesting the baseline model is stronger than randomization but can certainly be improved.



Week 6 Limitations

- Model is better at detecting skips than confirming non-skips
- Greedy sequencing is efficient but can miss globally better playlist orders
- Current objective still underweights transition smoothness and musical flow
- Probability calibration is not yet validated, so optimization may trust imperfect scores
- Results are offline only, so real-user results are still unverified
- Current skip-history features may be confounded by platform/user constraints (e.g., skip limits, session progression), so we will model Free vs Premium separately and add interaction terms

Next Steps: Strengthening Prediction Engine

Calibrate Probabilities

- Current model predicts skip more than non-skip

Can validate with ECE and Brier score.

- Platt scaling/isotonic regression

Want to improve trust in optimization objective

Improve non-skip detection

Address model's bias towards predicting skip

Want to rebalance loss and adjust threshold

Improve no-skip precision

Current sequencing too conservative, leads to skip over-prediction

Transition-aware features

Include pairwise song differences (tempo, key, energy, embeddings)

Model P(skip | previous song, candidate song, session)

Addresses underweighting of transition smoothness and musical flow

Static Prediction → Sequential Optimization

Replace Greedy with Beam Search

Evaluate top-K sequence candidates

Reduce risk of locally optimal but globally poor ordering

Explore Sequential Models (LSTM / RL)

Model session dynamics explicitly and predict skip probability conditioned on full history

Frame as a sequential decision problem

Validate with Counterfactual / Online Testing

Offline replay evaluation with A/B testing framework

Compare optimized vs baseline playlists